

BREECH-LOADING ARTILLERY

British engineer William Armstrong designed the first efficient breech-loading rifled field gun in 1855. The shell and gunpowder propellant were loaded at the breech, which was closed with a “vent-piece” secured in a slot with a hollow screw. Armstrong’s 12-pounder gun, of 1859, was the first rifled breech-loading field gun to enter British Army service, and the Armstrong RBL 40-pounder was an adaptation of this gun as a medium artillery piece. The French Canon de 75mm Modèle 1897 added a key element to artillery design – a recoil-dampening mechanism that kept the trail and wheels perfectly still when firing, which freed gun crews from having to re-aim the gun after each shot.



◀ **WHITWORTH 45MM BREECH-LOADING BOAT GUN**
Date 1875
Origin UK
Length 94cm (37in)
Calibre 45mm
Range 360m (393¼ yards)

This boat gun had hexagonal rifling with a Whitworth sliding-lock breech-loading mechanism. It was set on a cone mounting mostly used for small naval guns. This example was mounted on an armed yacht.



► **BL 15-POUNDER 7CWT**
Date 1892
Origin UK
Length 2.13m (7ft)
Calibre 76.2mm
Range 5.26km (3.26 miles)

This 15-pounder was fitted with an early recoil device – a spade that dug into the ground on firing, and was connected to a spring on the trail. The gun jumped backwards on firing and then forwards under the pressure of the spring. It could fire eight rounds per minute.



▲ **ARMSTRONG RBL 40-POUNDER GUN**
Date 1861
Origin UK
Length 3m (9¾ft)
Calibre 12cm
Range 2.56km (1.59 miles)

The Armstrong rifled breech-loading 40-pounder was used by the British Royal Navy as a broadside gun, and by the army as a defensive gun in military forts. It saw action in the Royal Navy’s bombardment of Kagoshima, Japan, in August 1863.



► **ARMSTRONG RBL 12-POUNDER**
Date 1859
Origin UK
Length 2.13m (7ft)
Calibre 7.62cm
Range 3.1km (1.92 miles)

The Armstrong rifled 12-pounder gun required a crew of nine men to operate it. The gun that entered British Army service in 1859 had a 2.13m (7ft) barrel, while the British Royal Navy used a 1.83m (6ft) barrel version. In 1863, the shorter version became standard.



► **HOTCHKISS QUICK-FIRING 3-POUNDER NAVAL GUN**
Date 1885
Origin France
Length (Barrel) 2m (6½ft)
Calibre 47mm
Range 3.66km (2.27 miles)

The Hotchkiss QF (Quick-Firing) 3-pounder was used by the British Royal Navy from 1885, as well as the French, Russian, and US navies. These guns were made by a division of the Armstrong armaments business. Operated by two men, the gun could fire up to thirty 1.5kg (3¼lb) steel shells per minute.